The Amplitude Law: the obstruction is the interpolation function, not the carrier—and a monotone kernel escapes it with a_0 untouched

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The acceleration scale $a_0 = \kappa c\sqrt{G\rho_\Lambda} = c^2\sqrt{\Lambda/32\pi} = 9.3619\times10^{-11}\,\mathrm{m\,s^{-2}}$ (with $\kappa=\frac{1}{2}$ *fitted*, not derived) reproduces the observed MOND scale from the cosmological constant alone, and the associated interpolation $g_{\rm obs}^2 = g_{\rm bar}^2 + a_0 g_{\rm bar}$ fits the SPARC radial acceleration relation to 0.108 dex. We ask what field theory, if any, makes the corresponding halo density $\rho = \sqrt{GM_b a_0}/(4\pi G r^2)$ a *dynamical consequence* rather than an initial condition. We first show that the equation-of-state route is not the binding constraint: in the weak field the anisotropy $p_r - p_t$ is unobservable (an exact flat direction of the two lensing observables), so the constraint is the amplitude, not the stress. We then exclude several carrier classes outright—every *static* single k-essence scalar obeys $p_t = -\rho$ identically for any kinetic function, and the general khronon fails by a scaling theorem because a_0 cannot be built from c and G alone. Of six candidate mechanisms subjected to adversarial verification, **none survives**. Four of them deliver the amplitude law exactly—and we show this is a weaker achievement than it appears: integrating Eq. (3) gives $v_c^4 = GM_b a_0$ with v_c exactly r-independent, so the amplitude law and "flat rotation curves at the baryonic Tully–Fisher value" are *one statement*, and all four mechanisms return the identical Bekenstein–Milgrom phantom density. The coefficient 1.000000 is returned by $\mu = x/(1+x)$ and $\mu = x/\sqrt{1+x^2}$ as well; it measures the deep-MOND normalisation and nothing else. What is genuinely established is narrower and real: the AQUAL functional is strictly convex, so the halo is a *unique* functional of ρ_b with zero free data and an attractor in radius. The decisive obstruction is *arm-level*: all four mechanisms reduce to $\nabla\cdot[(1-\mu_v/B^2)\nabla\Phi] = 4\pi G\rho_b$ for a general $\Phi(x,y,z)$ with no symmetry assumed, so **which field carries the halo cannot move the Cassini quadrupole—only the interpolation function can**. The two kernels previously considered fail at opposite ends of that function. We then answer the squeeze directly, on 175 SPARC rotation curves with $\Upsilon_\star$ refit per kernel: the standard monotone family $\mu_n(x) = x/(1+x^n)^{1/n}$ **clears it**. At $n=10$ the quadrupole is 0.08–0.21× the Cassini ceiling across the full $\pm2\sigma$ of the *measured* external field on both footings, and the 1 AU monopole is 4×10^{-28} of the Mars budget, against 5.6–6.4× and 3.3×10^4 for the interpolation of Eq. (2). **Crucially a_0 is untouched**: the deep-MOND limit is identical across the family to 5×10^{-7}, so Eq. (1), the amplitude law, the baryonic Tully–Fisher relation and the fitted κ all survive the swap. *The solar system was never a statement about a_0; it is a statement about the transition region.* The cost is stated rather than hidden: the RAR fit degrades from 0.108 to 0.127 dex. The binding obstruction therefore moves—it is no longer the solar system but the double count between the phantom and any clustering dark sector, which is kernel-independent and which no choice of μ_n affects.

I. THE SCALE AND THE QUESTION

The framework studied here takes a single input: that the MOND acceleration scale is set by the cosmological constant through

$$a_0 \;=\; \kappa\, c\sqrt{G\rho_\Lambda} \;=\; c^2\sqrt{\Lambda/32\pi}, \qquad \kappa = \tfrac{1}{2}, \tag{1}$$

giving $a_0 = 9.3619\times10^{-11}\,\mathrm{m\,s^{-2}}$ on the canonical footing ($\rho_{\rm DE}$, cH_Λ) and 1.1279×10^{-10} on an alternative footing ($\rho_{\rm total}$, cH_0). $\kappa=\frac{1}{2}$ **is fitted, not derived**; it is measured at 0.529 ± 0.034, and four candidate values lie within 2σ. Every number in this paper is quoted on both footings.

The framework's own interpolation—not Milgrom's—is

$$g_{\rm obs}^2 = g_{\rm bar}^2 + a_0\, g_{\rm bar} \iff \mu(x) = \frac{\sqrt{1+4x^2}-1}{2x}. \tag{2}$$

Its phenomenology is banked elsewhere: 0.108 dex on the SPARC RAR at $\Upsilon_\star = 0.70$, the baryonic Tully–Fisher relation, and a weak-lensing RAR fit from 40 kpc to 2.2 Mpc.

The question this paper asks is the one that separates a fitting formula from a theory. Equation (2) is equivalent to a halo of density

$$\rho(r) \;=\; \frac{\sqrt{GM_b a_0}}{4\pi G r^2} \tag{3}$$

in the deep regime—isothermal, locked to the *baryonic* mass, with a_0 setting the coefficient. *What makes Eq. (3) a consequence of field equations rather than something put in by hand?* We call this the **amplitude law**, and it is the whole of the closure problem.

II. THE EQUATION OF STATE IS NOT THE CONSTRAINT

It is natural to attack the problem through the halo's stress tensor, because lensing and dynamics respond to different combinations of it. With $\nabla^2\Psi = 4\pi G\rho$ and $\nabla^2\Phi = 4\pi G(\rho + p_r + 2p_t)$, agreement between lensing and dynamics requires $p_r + 2p_t = 0$.

This turns out to be a weak constraint, for two reasons. First, pressureless dust satisfies it trivially. Second, and more sharply, the map $(p_r, p_t) \mapsto (p_r + 2L,\ p_t - L)$ leaves *both* the dynamical and the lensing source unchanged for every L: with three stress unknowns and two observables there is an exact flat direction, and **the anisotropy $p_r - p_t$ is unobservable in the weak field**. The entire testable content is the single number $\epsilon \equiv (p_r + 2p_t)/\rho c^2$.

Calibrating ϵ against data: using the full published 15×15 covariance of the KiDS-1000 excess surface density, the tightest defensible bound is $|\epsilon| < 0.049$, while any non-relativistic carrier has $\epsilon = O(v^2/c^2) \leq 1.4 \times 10^{-4}$. The minimum headroom is 2.53 orders. The equation of state is therefore *not* what constrains this sector; the amplitude is.

III. WHAT IS EXCLUDED

Three classes are closed by theorems rather than by fits.

a. Static single k-essence. For $\mathcal{L} = F(X)$ with a static radial profile $\varphi(r)$ on any static spherically symmetric metric, both $T^t{}_t$ and $T^\theta{}_\theta$ reduce to $-F$, so

$$p_t = -\rho \quad \text{identically, for every } F. \tag{4}$$

The reason is structural: for static φ the gradient reaches the metric only through g^{rr}, so the reduced Lagrangian contains neither g_{tt} nor the areal radius, and the result follows for a *generic* reduced Lagrangian. Shift-symmetric Horndeski G_3 dies with it. Imposing $p_r = -2p_t$ then forces $F \propto X^{3/2}$—the deep-MOND AQUAL power appears unforced from a lensing requirement—but at $p_t/\rho c^2 = -1$, the wrong sign and 6.7 orders from any admissible value.

The escape is the framework's own: a shift-symmetric condensate $\varphi = Q_0 t + \psi(r)$ gives $p_t + \rho = -F' Q_0^2/g_{tt} \neq 0$, and the branch criterion is simply that the condensate flow be subluminal. **The theorem does not touch the framework's dark sector.** It does close every static AQUAL-type scalar as a stress carrier.

b. The radial-director class. Maxwell, general nonlinear electrodynamics, the at-rest khronon, the Einstein-aether vector and a spacelike director were each solved by exact metric variation. Four reach only an order-unity traceless anisotropy; the fifth reaches a small one by being exactly zero. A useful identity emerged: the required tangential pressure is the anomalous field energy density,

$$p_t(r) = \frac{g_{\rm obs}^2 - g_{\rm bar}^2}{8\pi G} = \frac{a_0 g_{\rm bar}}{8\pi G}. \tag{5}$$

c. The khronon, by scaling. The general khronon action contains no dimensionful parameter but G, so its static equations are invariant under $r \to \mu r$, $M \to \mu M$ and any induced density is homogeneous: $\rho_{\rm eff} \sim M^p r^{-2-p}$. The amplitude law needs the mass exponent $\frac{1}{2}$ *and* the radial exponent -2, i.e. $p = \frac{1}{2}$ and $p = 0$ simultaneously. **Incompatible.** Equivalently, a_0 is an acceleration and cannot be built from c and G alone. The bare exterior density is $\propto r^{-4}$, low by 11.7 orders at the MOND radius. The khronon nonetheless *screens* beautifully: the only static scalar available to it is $|{\rm d}\ln N|^2 = (g_{\rm total}/c^2)^2$, the total local field gradient, with contrast 3.3×10^7 between the Sun at 1 AU and the Galaxy at the Sun.

IV. THE DEFLATION: THE AMPLITUDE LAW IS THE FLAT ROTATION CURVE

Before grading any mechanism against Eq. (3), it must be said what that equation is worth. Integrating it gives $M_{\rm dark}(<r) = \sqrt{M_b a_0/G}\, r$, hence

$$v_c^2 = \frac{G M_{\rm dark}}{r} = \sqrt{G M_b a_0}, \qquad \frac{\partial v_c^2}{\partial r} = 0 \quad \text{exactly}, \tag{6}$$

so $v_c^4 = G M_b a_0$: the baryonic Tully–Fisher relation. The converse also holds exactly. **The "amplitude law" and "flat rotation curves at the BTFR value" are one statement, not two.** Rewriting the interpolation as a density is a change of variables, not a derivation, and asking what makes Eq. (3) an attractor is asking what makes rotation curves flat—the original question, not a sharper one.

Two consequences follow, and both cut against the mechanisms below. First, *four* of the six deliver the amplitude law, and all four return the *identical* function of r, equal to the Bekenstein–Milgrom phantom density of Eq. (2): the ratio to Eq. (3) is 0.4472/0.7071/0.9487/0.9950/0.99995 at 0.5/1/3/10/100 r_M, footing-independent. Four successes are one success counted four times. Second, the coefficient is not diagnostic: $\mu = x/(1+x)$ and $\mu = x/\sqrt{1+x^2}$ return exactly 1 as well, and a kernel with deep limit $A\sqrt{a_0 g_b}$ returns exactly A. The "1.000000" measures the deep-MOND normalisation, which every MOND kernel shares. **It is not evidence for Eq. (2) over any other interpolation.**

What is left is narrower and genuinely a theorem. Since ${\rm d}(\mu x)/{\rm d}x = 2x/\sqrt{1+4x^2} > 0$ on $(0, \infty)$, the AQUAL functional is strictly convex, the elliptic problem has a unique solution, and the integration constant is killed by regularity rather than fitted. **The halo is a unique functional of ρ_b with zero free data**, and it is an attractor in radius: forward integration from inner data wrong by +50%, −30%, or a 10× spurious dark point mass converges to the same coefficient by 100 r_M.

V. SIX MECHANISMS, NO SURVIVOR

Six mechanisms were constructed and each subjected to independent adversarial verification on separate lenses (amplitude, screening, theoretical health). **None survived.** Two were dead on the amplitude law itself. The remaining four each fell on a health or screening lens: an AQUAL conformal coupling on $c_T \neq 1$ by 7.3 orders and on force screening that delivers an *enhancement* where suppression is required; a khronon on gravitational Cherenkov by 14–16 orders, with its anomalous acceleration identical at every planet to nine significant figures over a 6000× range in g_N; a mimetic construction on $c_s^2 \equiv 0$ forced by its own closure, hence Hadamard ill-posedness; and a two-field lock on a parallel-mode ghost together with the quadrupole of Sec. VI. *That last refutation is contested on a specific ground*: it covariantised the mediator as a gauge vector, whereas the aether-scalar-tensor vector is Lagrange-multiplier-constrained and unit-timelike, which removes the mode carrying the ghost. Its honest status is refuted-as-covariantised, and the deciding recomputation is named in Sec. VII and not performed here.

The two-field mechanism, which went furthest, carries $\Omega_{\rm dm}$ in a dust sector χ and locks it to the baryons through a mediator A with free function μ_v. Its content is: Gauss's law for the mediator, Gauss's law for gravity, and a pointwise balance between them. Combining these eliminates A and gives, **for any μ_v and with no interpolation function used at all**,

$$\boxed{\frac{GM_\chi(r)}{r^2} \;=\; g_{\rm obs} - g_{\rm bar}} \tag{7}$$

as an identity. *The real dark mass equals the phantom mass as a theorem, not an assumption.* This is the central result of the paper, and it is kernel-independent.

Feeding Eq. (2) into Eq. (7) gives $M_\chi = M_b(\sqrt{1+u^2} - 1)$ with $u = g_{\rm bar}/a_0$, hence

$$\rho_\chi(r) \;=\; \frac{r}{\sqrt{r^2 + r_M^2}}\,\frac{\sqrt{GM_b a_0}}{4\pi G r^2}, \qquad r_M = \sqrt{GM_b/a_0}. \tag{8}$$

An independent `brentq` plus finite-difference solve on a 200,000-point grid returns log-slope −1.99982, coefficient 0.9999975 of $\sqrt{GM_b a_0}/(4\pi G)$, and $\mathrm{d}\ln(\mathrm{coef})/\mathrm{d}\ln M_b = 0.500000$ over 10^9–$10^{12}\, M_\odot$. A pure-number trap test rejects factors of 2, $\frac{1}{2}$, 4π and $\sqrt{2}$. On a Hernquist source $\rho_\chi > 0$ at 0 of 400,000 radii, so it is not a point-mass artefact. The $1/r^3$ trap that kills k-essence is avoided structurally: μ_v tends to a constant in the deep regime, so the mediator is Maxwell-like there and holds only $\sim 2\times 10^{-7}$ of the dust's energy.

Screening is structural rather than tuned: μ_v's argument is $|\nabla A|$, and the balance makes $|\nabla A| = |\nabla\Phi|/B$ pointwise, so the free function eats the local *total* field gradient—the one quantity with the required dynamic range. The weak-contrast alternatives (potential 1.51×, dark density 2.22×, dispersion 1.00×) are untouched by the construction.

The mediator's Gauss law caps the dark field at $GM_\chi/r^2 \le C_\nu a_0$ with $C_\nu = 0.5$ exactly for Eq. (2). This is not an extra restriction: $\sup_{g_{\rm bar}}(g_{\rm obs} - g_{\rm bar}) = a_0/2$ exactly for the same interpolation, approached monotonically from below. The cap is the interpolation restating its own supremum. It does *not* forbid clusters: a calibrated cluster ($M_{500} = 5\times 10^{14} M_\odot$, $R_{500} = 1.3$ Mpc, $f_{\rm bar} = 0.13$, giving $\eta = 1.79$ canonical / 1.64 alt) requires a dark field of $0.383a_0$ / $0.318a_0$, which is 1.30× / 1.57× inside the cap.

VI. THE OBSTRUCTION

No mechanism closes the theory, and the reason is not specific to any of them. Applying `sympy` to a general $\Phi(x,y,z)$ with no symmetry assumed gives residual exactly zero for

$$\nabla\cdot\left[\left(1 - \mu_v/B^2\right)\nabla\Phi\right] = 4\pi G\rho_b, \tag{9}$$

and the same reduction holds for the conformal, khronon and relaxation constructions. **All four share one baryon sector.** Consequently the statement "baryons carry no dark charge, so there is no fifth force" is true about field content and observationally vacuous, and—decisively—**which field carries the halo cannot move the Cassini external-field quadrupole Q_2. Only the interpolation function can.** The obstruction is arm-level, not mechanism-level, which is why six structurally different constructions all died in the same place.

Against the published ceiling $Q_2 \le 5.2\times 10^{-27}\,\mathrm{s}^{-2}$ (2σ), three referees on three different mechanisms returned $Q_2 = (1.5$–$4.6)\times 10^{-26}\,\mathrm{s}^{-2}$, i.e. 2.9–8.9× the ceiling, $+7.6\sigma$ to $+24.9\sigma$. *Against this finding, and stated because it matters:* the published spread of this test is 3–15σ depending on the Milky Way mass model. The honest floor is $\gtrsim 3\sigma$, or $> 1.5\times$ the ceiling at every corner examined. A real failure, but not a clean single-number kill.

The two candidate kernels fail at *opposite ends of the same function*, which is the shape of the remaining problem. Equation (2) has an anomalous sunward acceleration saturating at exactly $a_0/2$, exceeding the Mars ephemeris budget by 3.34×10^4 (canonical) / 4.03×10^4 (alt). *This corrects an earlier figure in this programme's own records by a factor ∼27, in the adverse direction; two independent derivations agree.* An exponential kernel is monopole-void at 1 AU (10^{-3459}) but returns a quadrupole 1.40× *worse*, because Q_2 is sourced at $y \approx 2$ while 1 AU sits at $y = 6.3\times 10^7$—seven orders above the region that sets it, so the exponential tail is real and is being evaluated in the wrong place.

Finally, the interpolation's ephemeris liability and its realisability are the same algebraic fact: in two-potential form the free function is $\mu_s(s) = k\,s/(1-2s)$ with a simple pole at $s = \frac{1}{2}$, and that pole is what makes Eq. (2) realisable as a coupled scalar at all. The bug and the theory's existence cannot be separated.

VII. THE DOUBLE COUNT, AND WHY IDENTIFYING THE FIELDS DOES NOT FIX IT

Equation (9)'s phantom fills the entire discrepancy, so solving $M_b + M_{\rm cond} + M_{\rm ph} = M_b\nu$ returns $M_{\rm cond} = 0$ identically. Any dark sector that also clusters is pure overshoot: at the cosmic share $\Omega_{\rm dm}/\Omega_b = 5.375$ it exceeds the observed curve by 32.5/25.7/11.5/3.6× at 0.5/1/3/10 r_M, fatal inside 443 kpc (canonical) / 403 kpc (alt), and roomy only beyond.

The natural escape is that the AQUAL scalar and the condensate are one field. They are; but it does not help. The cross term that would cancel the double count is, as a theorem,

$$\frac{\rm cross}{\rm dust} = -\frac{\kappa^2 s^3}{24\pi}, \qquad s \equiv |\nabla\varphi|/a_0, \tag{10}$$

a *pure number* of s—the mass scale, the brane tension, the charge, G and even the local a_0 all cancel. Cancellation requires $s^\star = 6.7062$, while Eq. (2) caps $s \le \frac{1}{2}$: short 13.4× in gradient and 2413× in energy. The structural reason is that the MOND phantom is not a stress and cannot be, so the anomalous force lives in a non-$T_{\mu\nu}$ channel while the condensate energy gravitates through $T_{\mu\nu}$. They are different *channels* of one field, which is exactly why merging the fields does not merge the contributions.

The aether-scalar-tensor theory avoids this, but by an ansatz rather than a mechanism: its quasi-static limit leaves the scalar's tick unperturbed, capping halo overdensity at 20.9× the cosmic mean where an isothermal halo at 20 kpc needs 5.5×10^4. Its galaxy solutions therefore contain no clustered dark sector at all. The double count is absent in the regime that theory solves, and that theory does not solve the regime where it would appear—nonlinear structure formation there is unsolved, and remains so here.

VIII. THE MONOTONE ESCAPE

The preceding section leaves a single question, and because the obstruction is arm-level it is a question about a *function*, not about a mechanism: does any interpolation fit the RAR, satisfy the quadrupole bound, and keep the 1 AU monopole inside ephemeris budgets? We answer it on 175 SPARC rotation curves with $\Upsilon_\star$ refit per kernel and the AQUAL (not QUMOND) quadrupole, which is the conservative choice.

The standard published family

$$\mu_n(x) = \frac{x}{(1+x^n)^{1/n}} \tag{11}$$

clears it. The family is monotone—$\mathrm{d}\mu/\mathrm{d}x = (1+x^n)^{-1-1/n} > 0$—so the AQUAL functional remains strictly convex and the uniqueness result of Sec. IV survives the swap intact.

μ_{10} clears across the full $\pm2\sigma$ of the *measured* external field on both footings (0.050–0.351× the ceiling); μ_5 clears everywhere on the canonical footing.

a. a_0 is untouched, and this is the point. The deep-MOND limit is identical for every member of the family to 5×10^{-7} at $y = 10^{-12}$. Equation (1), the amplitude law, the BTFR and the κ measured from it therefore survive the kernel swap unchanged. *The solar-system liability was never a statement about a_0; it was a statement about the transition region, and the transition region is exactly where the interpolation is free.*

b. The cost. Equation (2) remains the better RAR fit: rms 0.108 against 0.127 dex, χ^2/dof 21.1 against 49.1. The kernel that survives Cassini fits the rotation curves worse. That trade is the result, and we decline to report one half of it without the other.

c. A correction against an earlier claim. A no-go asserted at an intermediate stage of this work—that every monotone kernel carries $Q_2 \ge 3.5\times$ the ceiling—does not reproduce and is withdrawn. Its error ran in the direction of manufacturing a deficit. Separately, a parallel-mode ghost attributed to the two-field mechanism does not exist on Eq. (2): the relevant kinetic coefficient is $1 - 2x/\sqrt{1+4x^2} > 0$ for every real x, with no root, and $c_T = 1$ exactly.

IX. WHERE THE OBSTRUCTION ACTUALLY IS

With the solar system cleared by a kernel choice, the binding constraint is the one that no kernel touches. The phantom of Eq. (9) fills the entire discrepancy, so any sector that *also* clusters is pure overshoot—and per Sec. VIII identifying that sector with the scalar does not help, because the cancelling cross term is capped at $\kappa^2/(192\pi)$ while cancellation would require a field gradient 13.4× beyond a hard bound. This failure is kernel-independent: it is untouched by every member of the μ_n family, and it is now the wall.

The calculation that decides the programme is therefore no longer the squeeze. It is: *what carries $\Omega_{\rm dm}$ in the cosmological sector without depositing a clustered halo in galaxies?* Two classes remain unevaluated and are the

TABLE I. 175 SPARC curves, $\Upsilon_\star$ refit per kernel, AQUAL quadrupole. Q_2 is quoted as a multiple of the 2σ Cassini ceiling, canonical / alt footing; the monopole as a multiple of the Mars ephemeris budget.

kernel	$\Upsilon_\star$	RAR rms	Q_2/ceiling	monopole
exponential	0.62	0.0998	7.77 / 8.52	0
Eq. (2)	0.70	0.1083	5.59 / 6.39	3.3×10^4
μ_3	0.81	0.1179	1.55 / 2.44	2.8
μ_5	0.84	0.1233	**0.39 / 0.82**	2.7×10^{-8}
μ_{10}	0.85	0.1266	**0.08 / 0.21**	4×10^{-28}

natural place to look—Vainshtein and k-mouflage screening, the only class that suppresses the force rather than the information, which this work did not run at all; and multi-streaming support, which would supply the halo with no second field and no modified Poisson equation, and would therefore be immune to both obstructions at once.

X. WHAT IS NOT CLAIMED

1. **No mechanism survived.** An earlier version of this paper reported that one of six did. That is withdrawn: the mechanism in question carried a third referee, on the theoretical-health lens, which refuted it. Its refutation is in turn contested on the ground given in Sec. V, and the deciding recomputation—redoing the covariantisation with a Lagrange-multiplier-constrained unit-timelike vector rather than a gauge vector—has not been done.

2. **Delivering the amplitude law is not evidence for this framework.** Per Sec. IV it is equivalent to flat rotation curves at the BTFR value, four mechanisms return the identical Bekenstein–Milgrom density, and the coefficient is shared by every MOND kernel.

3. **The cosmological leg is untested**, and **ghost-freedom and $c_T = 1$ are uncomputed** for the constrained-vector realisation.

4. $\kappa = \frac{1}{2}$ **remains fitted**, and nothing here derives it.

5. **Clusters remain short by** $\approx 2\times$, inherited unchanged.

6. **Untried, and named because their absence is a limitation of this work:** emergent/entropic gravity—the one published derivation of exactly this amplitude law, whose scale $cH_0/6 = 1.0914 \times 10^{-10}\,\mathrm{m\,s^{-2}}$ sits within 3.3% of the alternative footing; nonlocal gravity; Vainshtein/k-mouflage screening, which is the one class that suppresses the *force* rather than the information; multi-streaming caustics as the source of the support; and QUMOND. None was evaluated.

XI. WHAT WOULD DECIDE IT

Because the obstruction is arm-level, the deciding calculation is not another mechanism. It is a question about a function:

> *Does there exist an interpolation $\nu(y)$ that simultaneously (a) fits the SPARC RAR at ≤ 0.06 dex intrinsic with $\Upsilon_\star$ inside the Spitzer prior, (b) gives $Q_2 \leq 5.2 \times 10^{-27}\,\mathrm{s^{-2}}$ at $g_{\rm ext} = 1.9$–$2.6\,a_0$ computed with the AQUAL rather than QUMOND quadrupole, and (c) keeps the 1 AU monopole inside the per-planet ephemeris budgets?*

If such a ν exists, the modified-Poisson arm survives and the mechanism question reopens. If none exists, the arm is closed by a no-go, and only constructions that escape the modified-Poisson framing entirely—entropic, nonlocal, or multi-streaming—remain available. We do not know which world this is, and we regard establishing it as more valuable than any further mechanism search.

XII. REPRODUCIBILITY

Every load-bearing claim is backed by a committed script that exits zero. The results above were produced under adversarial verification in which independent agents were instructed to refute each claim, with a standing rule that manufacturing a deficit is penalised as heavily as manufacturing a win. Errors were caught in both directions during this work and are logged in the repository, including one case where a first draft asserted an obstruction was fatal at every radius while its own table showed otherwise.

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